TUTORIUM - II

Def: (Skalarfeld)

$$f : \mathbb{R}^m \longrightarrow \mathbb{R}$$

Def: (Vektorfeld)

$$F : \mathbb{R}^m \longrightarrow \mathbb{R}^n$$

Bem: (vgl. Z2.1)

Sei $\gamma : \mathbb{R} \longrightarrow \mathbb{R}^3$

$$K(t) = \frac{\| \dot\gamma(t) \times \ddot\gamma(t) \|}{\| \dot\gamma(t) \|^3}$$

$$s(t) = \int_{t_0}^{t_1} |\dot\gamma(t)| \, dt$$

$\Rightarrow$ KURVENINTEGRAL VOM TYP I:

$f : \mathbb{R}^m \to \mathbb{R}$, $\gamma : \mathbb{R} \to \mathbb{R}^m, t \mapsto \gamma(t)$, $t \in [t_0, t_1]$

$$\int_\gamma f(x)\, dx = \int_{t_0}^{t_1} |f(\gamma(t))| \, |\dot\gamma(t)| \, dt$$

$\Rightarrow$ Kurvenintegral vom Typ II:

$F : \mathbb{R}^m \to \mathbb{R}^n$, γ: $\qquad F \in C^1$

$$\int_\gamma F(x) \cdot dx = \int_{t_0}^{t_1} \langle F(\gamma(t)), \dot\gamma(t) \rangle \, dt$$

$\phi : \mathbb{R}^3 \to \mathbb{R}$ | $\vec{A} : \mathbb{R}^3 \to \mathbb{R}^3$

$\vec{E} = -\operatorname{grad} \phi$ | $\vec{B} = \operatorname{rot} \vec{A} = \vec\nabla \times \vec{A}$

P2.1: $\Gamma(t) = \begin{pmatrix} \gamma_1(t) \\ \gamma_2(t) \\ 0 \end{pmatrix}$

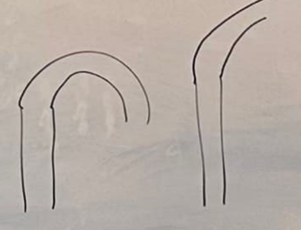
$$\begin{cases} F(t) = \int_0^t \underbrace{\cos\left(\frac{u^2}{2}\right)}_{f(u)} du \\ F'(t) = f(t) \end{cases}$$

$$\dot\Gamma(t) = \begin{pmatrix} \frac{d}{dt} \int_0^t \cos\left(\frac{u^2}{2}\right) du \\ \frac{d}{dt} \int_0^t \sin\left(\frac{u^2}{2}\right) du \\ 0 \end{pmatrix} \overset{HDI}{=} \begin{pmatrix} \cos\left(\frac{t^2}{2}\right) \\ \sin\left(\frac{t^2}{2}\right) \\ 0 \end{pmatrix} \Rightarrow \ddot\Gamma(t) = t \begin{pmatrix} -\sin\left(\frac{t^2}{2}\right) \\ \cos\left(\frac{t^2}{2}\right) \\ 0 \end{pmatrix}$$

$$\left\| \begin{pmatrix} \dot\gamma_1 \\ \dot\gamma_2 \\ 0 \end{pmatrix} \times \begin{pmatrix} \ddot\gamma_1 \\ \ddot\gamma_2 \\ 0 \end{pmatrix} \right\| = \sqrt{(\dot\gamma_1 \ddot\gamma_2 - \ddot\gamma_1 \dot\gamma_2)^2}$$

$$K(t) = \frac{t\left(\cos^2\left(\frac{t^2}{2}\right) + \sin^2\left(\frac{t^2}{2}\right)\right)}{\left(\cos^2\left(\frac{t^2}{2}\right) + \sin^2\left(\frac{t^2}{2}\right)\right)^{3/2}} = t$$

$$S(t) = \int_0^t \underbrace{\sqrt{\cos^2 \frac{t^2}{2} + \sin^2 \frac{t^2}{2}}}_{=1} \, dt = t$$



P2.2: $\gamma(t) = \begin{pmatrix} t\cos t \\ t\sin t \\ \frac{2\sqrt{2}}{3} t^{3/2} \end{pmatrix}$, $t \in [0,1]$

$\dot\gamma(t) = \begin{pmatrix} \cos t - t\sin t \\ \sin t + t\cos t \\ \sqrt{2\,t} \end{pmatrix}$

$\lambda(x,y,z) = z^2 = \tilde\lambda(z)$

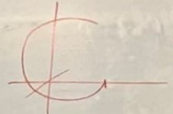
$|\dot\gamma(t)| = \sqrt{\boxed{\cos^2 t + \sin^2 t} + \boxed{t^2\sin^2 t + t^2\cos^2 t} - 2t\cos t\sin t + 2t\cos t\sin t + 2t} = \sqrt{1+t^2+2t} = 1+t$

$$M = \int_\gamma \lambda(x,y,z)\, d(x,y,z) = \int_\gamma \tilde\lambda(z)\, dz$$

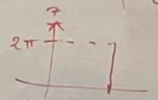
$$= \int_0^1 |\lambda(\gamma(t))|\,|\dot\gamma(t)|\,dt = \int_0^1 \left(\frac{8}{9} t^3\right)\cdot(1+t)\,dt$$

$$= \frac{8}{9}\left[\frac{1}{4} + \frac{1}{5}\right] = \frac{8}{20} = \frac{2}{5}$$

P2.3: $\gamma_1(t) = \begin{pmatrix} \cos t \\ \sin t \\ t \end{pmatrix}$, $t \in [0,2\pi)$ $\quad \gamma_2(t) = \begin{pmatrix} 1 \\ 0 \\ t \end{pmatrix}$, $t \in [0,2\pi)$



$F(x,y,z) = \begin{pmatrix} -y \\ x \\ z \end{pmatrix}$



$\boxed{\tilde\gamma_2(t) = \begin{pmatrix} 1 \\ 0 \\ 4\pi - t \end{pmatrix}, \ t \in [2\pi, 4\pi]}$

$\dot\gamma_1(t) = \begin{pmatrix} -\sin t \\ \cos t \\ 1 \end{pmatrix}$; $\quad F(\gamma_1(t)) = \begin{pmatrix} -\sin t \\ \cos t \\ t \end{pmatrix}$

$$\int_0^{2\pi} \left\langle \begin{pmatrix} -\sin t \\ \cos t \\ 1 \end{pmatrix}, \begin{pmatrix} -\sin t \\ \cos t \\ t \end{pmatrix} \right\rangle dt = \int_0^{2\pi} (1+t)\,dt$$

$\dot\gamma_2(t) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ $\quad F(\gamma_2(t)) = \begin{pmatrix} 0 \\ 1 \\ t \end{pmatrix}$

$$\int_0^{2\pi} \left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ t \end{pmatrix} \right\rangle dt = \int_0^{2\pi} t\,dt$$

$$\int_{\gamma_1} F(x)\cdot dx - \int_{\gamma_2} F(x)\cdot dx = \int_0^{2\pi} (1+t)\,dt - \int_0^{2\pi} t\,dt = \int_0^{2\pi} (1 + \cancel{t} - \cancel{t})\,dt$$

$$= 2\pi$$